Glossa Lab -- Research Brief for Dr. Andreas Fuls

May 2026 -- Structural Analysis of the Indus Script

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Attachment to ICIT access request email.

1. Project overview

Glossa Lab is an open-source computational platform for structural analysis and hypothesis-driven decipherment of undeciphered writing systems. Available at:

Core methodology: **compression as structure discovery.** Entropy profiles, spectral analysis, and Zipf statistics are used as compression metrics -- the goal is to minimize the description length of the corpus, which reveals the latent grammar that makes sequences non-random. Only after structural constraints are established are linguistic hypotheses tested, following an explicit anti-circularity protocol.

Technology stack: FastAPI (Python), React/TypeScript, SQLite, graph-based experiment executor. Every experiment is version-controlled and registered in a structured job ledger with reproducible JSON output.

2. Data sources

Source	Inscriptions	Sign numbering	Coverage
Holdat LLC (2025)	1,670 seals	Mahadevan M-numbers	9 sites: Mohenjo-daro, Harappa, Dholavira, Chanhu-daro, Kalibangan, Lothal, Surkotada, Amri, Rakhigarhi
Mahadevan 1977 (M77)	1,669 inscriptions	M-numbers	Full concordance, 390 distinct signs
Mahadevan 2003 (TB)	47-110 inscriptions	Aksharas	Tamil-Brahmi corpus, Harvard Oriental Series 62
Parpola 2010	--	P-numbers	Iconographic anchors, Dravidian phoneme hypothesis
DEDR (Burrow & Emeneau 1984)	--	--	Dravidian Etymological Dictionary

Note on corpus need: The ICIT corpus (Fuls 2023, 4,537 artefacts) would expand the primary data source by 2.7x and allow testing whether the correlations below hold at scale. This is the most important outstanding dependency.

3. Core structural findings (Phase-30/31, May 2026)

3.1 Spectral gap universality

A length-stratified spectral analysis of the M77 corpus across 8 inscription length bins (L1-L9+) shows **spectral_gap = 0.0 in every stratum**:

Bin	Inscriptions	spectral_gap	Verdict
L1-1	564	0.0	Maximally deterministic
L2-2	357	0.0	Maximally deterministic
L3-3	270	0.0	Maximally deterministic
L4-4	188	0.0	Maximally deterministic
L5-5	112	0.0	Maximally deterministic
L6-6	73	0.0	Maximally deterministic
L7-8	61	0.0	Maximally deterministic
L9+	44	0.0	Maximally deterministic

All eigenvalues cluster at 1.0. This falsifies the "short-inscription noise" hypothesis: the structural anomaly is not an artifact of short sequences; it is corpus-wide. This is the expected behavior of a writing system with a consistent positional grammar.

3.2 Zipf slope comparison (Phase-31 T3) [VERIFIED]

Corpus	Zipf slope	Regime		
Indus M77	0.75	Syllabic / logo-syllabic (0.5-1.5)		
Tamil-Brahmi	0.93	Syllabic / logo-syllabic (0.5-1.5)		
	delta		**0.18**	Within threshold 0.3

Both corpora fall in the recognized syllabic power-law regime. The pre-registered threshold of 0.3 was set based on the distribution of known script pairs. At delta=0.18, the test is FAVORABLE for Dravidian script-class alignment.

This is the cleanest result: it does not depend on phoneme assignments and is not circular. [VERIFIED -- Phase-31]

3.3 Tamil-Brahmi phoneme correlation (V24) [VERIFIED]

After 17 autonomous distributional assignment rounds on the Holdat corpus:

Metric	Value
Signs assigned	333 / 390 (85.4%)
Token coverage	99.2%
Fully decoded inscriptions	96.7% (1,615 / 1,670)
Weighted confidence	64.8% (HIGHx1.0, MEDIUMx0.6, LOWx0.3)
Confidence breakdown	HIGH: 9 · MEDIUM: 63 · LOW: 261
TB phoneme correlation	**0.907** Pearson r
Random baseline (1,000 permutations)	0.470
Percentile rank	100th ($p < 0.001$)

The correlation of 0.907 measures alignment between the phoneme frequency distribution implied by the current sign assignments and the Tamil-Brahmi phoneme frequency distribution from Mahadevan 2003. This is a structural alignment metric, not a claimed phonetic identity. The gap between 0.907 (achieved) and 0.470 (random baseline) is large and statistically robust.

Epistemic caveat (mandatory): The 333 assignments are distributional hypotheses, NOT verified phonetic readings. They are not cross-validated against bilingual material. Publication-grade validation requires: (1) ICIT corpus, (2) Gulf-type seal cross-reference, (3) independent bilingual or semi-bilingual material.

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4. Phase-29d Enmenanak finding [INFERRED -- statistically robust]

A reverse Janabiyah search across 1,222 Sumerian personal names from ePSD2 identified **Enmenanak** as the top candidate for the Janabiyah Boss inscription pattern:

Test	Result
Score	7.0 (100th percentile of null, $p < 0.001$)
A1: Permutation test	SIGNIFICANT -- score > 95th percentile
A2: Period filter (Ur III / Old Akkadian)	FAVORABLE -- 4 candidates survive
A3: Meluhha co-occurrence	NEUTRAL -- no direct evidence (does not falsify)
Overall	**SURVIVES A1+A2+A3 -- statistically robust**

Status: [INFERRED], pending ICIT corpus cross-validation.

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5. Phase-32 T4 SA decipherment (word-level LM) [NEUTRAL]

A simulated annealing run against a clean Dravidian Tamil bigram LM (DEDR + Sangam corpus, 486 bigrams, zero English contamination) produced:

? mean_consistency: 0.297 (low convergence across 5 seeds)

? hci_count: 4 (only 4 signs robustly assigned across seeds)

? Verdict: NEUTRAL / INCONCLUSIVE

Root cause: 486 bigrams is too sparse for the SA to discriminate Dravidian word-pair patterns from random. The LM is insufficient for this test, not the hypothesis.
The V24 TB correlation (0.907) and Phase-31 T3 Zipf result (delta=0.18) are not affected.

6. Anchor set

The current 9 HIGH-confidence anchors:

Sign	Reading	Source
M047	min (fish)	Parpola iconographic + crosswalk
M087	vel	Numeral + DEDR
M088	mu(n)-	Numeral + DEDR
M091	aru-	Numeral + DEDR
M092	elu-	Numeral + DEDR
M086	or-	Numeral + DEDR
M175	katir	Iconographic + DEDR
M261	muruku	Iconographic + Parpola
M281	pillai	Iconographic + Parpola

M267 is classified as UNCERTAIN (high frequency on all motif types; cannot be a semantically specific reading). M099 is confirmed terminal (avg_position 0.598, is_ending=True; reading kol as Dravidian auxiliary is positionally consistent).

7. Outstanding dependencies

Item	Status	Action
ICIT corpus (4,537 artefacts)	Pending Dr. Fuls	This email
Gulf-type seal cross-reference (Laursen Table 1, 23 objects)	Sources identified	Awaiting ICIT for ECIT crosswalk
Phase-32 T1: TB-NAMES extraction	Not started	Mahadevan 2003 proper names
Phase-32 T2: TB parser coverage 100+/110	Not started	Requires epub access
Phase-32 T5: ICIT re-run	Blocked	Awaiting ICIT access

8. Anti-circularity statement

The following anti-circularity measures are explicitly implemented:

1. **Structural analysis precedes phonetic assignment** -- Zipf, spectral, positional, and

bigram analyses are computed before any phoneme hypotheses are tested.

2. **LM is built from independent sources** -- the Dravidian Tamil LM uses DEDR root forms and Sangam corpus texts, not Indus inscriptions.

3. **High-positional-bias signs are excluded from anchor assignments** -- signs with positional bias > 0.9 are not used as phonetic anchors.

4. **All results carry explicit epistemic markers** -- VERIFIED, INFERRED, ASSUMPTION, UNCERTAIN, UNCERTAIN are used per the AEE (Applied Epistemic Engineering) protocol.

5. **Null distributions are computed before claiming significance** -- all percentage-rank claims are backed by explicit permutation tests (>= 1,000 iterations).

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9. Contact and repository

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All results are reproducible. The repository contains the full experiment ledger, JSON output files, and source code under open-source license.

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